Improving Galaxy Morphology Classification under Class Imbalance with VAE-Based Generative Augmentation

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***Abstract*—Large imaging surveys produce far more galaxy images than can be labelled manually, motivating automated morphological classification with deep learning. Public benchmarks such as Galaxy10 DECaLS remain class-imbalanced, which can degrade recognition of rare morphologies. This paper investigates whether synthetic images from a variational autoencoder (VAE) improve convolutional neural network (CNN) classification under that imbalance. Following the CRISP-DM methodology, we train a convolutional VAE on Galaxy10 DECaLS training images, sample class-conditional latent codes to balance under-represented morphologies, and compare ResNet50 transfer-learning classifiers trained on real images only versus real images plus VAE synthetics. On a held-out test set of 2,661 galaxies, generative augmentation raises overall accuracy from 67.08% to 71.51% and macro-F1 from 0.648 to 0.686. For the minority *Cigar Shaped Smooth* class, recall increases (0.62 to 0.64) while F1 decreases slightly (0.608 to 0.571) due to lower precision. The results support VAE augmentation as an effective global regulariser under imbalance, while highlighting a precision–recall trade-off on the scarcest class. Code and configuration are released with the submission for reproducibility.**

***Index Terms*—galaxy morphology, convolutional neural networks, variational autoencoders, class imbalance, generative data augmentation, transfer learning**

I. Introduction

Modern sky surveys deliver imaging volumes that exceed the capacity of purely manual morphological labelling [1], [2]. Citizen-science projects such as Galaxy Zoo provide invaluable labels, yet rare morphological types remain scarce, creating class imbalance that challenges supervised deep learning [6], [9]. Morphology remains scientifically important because it correlates with star-formation history, merger activity and environment; therefore errors concentrated on rare classes can bias demographic inferences even when overall accuracy looks acceptable.

Building on the long lineage of neural pattern recognisers [16], [17], convolutional networks are now standard for galaxy morphology [5], and transfer learning from ImageNet-pretrained backbones is a practical recipe taught in deep-learning curricula [6], [10]. In parallel, generative models—especially variational autoencoders (VAEs) [7]—learn latent representations from which new samples can be drawn [8], [9]. Generative augmentation is therefore a natural candidate for mitigating imbalance, but its benefit for astronomical morphology is not automatic: synthetic fidelity, label noise, and evaluation protocol all matter. In particular, a generative model that improves reconstruction loss may still harm a downstream classifier if synthesised samples occupy ambiguous regions of pixel space.

Class imbalance interacts with deep networks in several ways. Gradient updates are dominated by frequent classes; decision thresholds calibrated on aggregate accuracy favour majority labels; and minority recall can collapse without the practitioner noticing if only top-line accuracy is monitored [9]. Classical remedies include re-weighting, oversampling and generic image augmentation strategies [14]. Generative oversampling differs because new minority examples are not literal copies: a VAE can, in principle, interpolate morphological factors encoded in z. Whether those factors are faithful for thin "cigar-shaped" galaxies is an empirical question addressed here.

This study asks:

> *Given class imbalance in Galaxy10 DECaLS, do VAE-generated synthetic images improve CNN morphological classification on minority classes without degrading majority-class performance?*

We answer the question empirically with a controlled A/B design under CRISP-DM [11]: (A) ResNet50 trained on real images only; (B) the same architecture trained on real images plus class-balancing VAE synthetics. Both models are evaluated on an identical stratified held-out test set. The main contributions are: (i) a reproducible CRISP-DM pipeline combining a ConvVAE and ResNet50 transfer learning on Galaxy10 DECaLS; (ii) quantitative evidence that VAE augmentation improves global accuracy and macro-F1; and (iii) a critical analysis of the minority-class precision–recall trade-off.

The remainder of this paper reviews related work (Section II), details the methodology (Section III), presents evaluation results (Section IV), discusses reproducibility and threats (Sections V–VI), and concludes with limitations and future work (Section VII).

II. Related Work

A. Galaxy morphology and Galaxy10

Galaxy Zoo demonstrated that morphological labels can be crowdsourced at scale [1], [2]. Early Galaxy Zoo papers already noted that fine morphological distinctions are harder for volunteers than coarse spiral/elliptical splits, foreshadowing class-dependent label noise. Galaxy10 DECaLS packages 17,736 colour cutouts into ten broad morphology classes with labels derived from Galaxy Zoo and images from DESI Legacy Imaging Surveys [3], [4], [15]. It is a convenient supervised benchmark, but class counts are uneven (e.g., *Cigar Shaped Smooth* is markedly under-represented), so methods that ignore imbalance may under-serve rare types. Compared with raw Galaxy Zoo vote fractions, Galaxy10's discrete ten-class formulation simplifies supervised learning at the cost of discarding vote uncertainty—another reason to treat hard-label metrics cautiously.

B. CNNs for astronomical images

Early deep morphological classifiers established that CNNs outperform classical feature pipelines on Galaxy Zoo-like tasks [5]. Dieleman et al. further exploited rotational symmetries, underscoring that inductive bias matters in astronomy where orientation is often nuisance variation. Residual networks [6] and transfer learning further reduce labelled-data requirements by reusing ImageNet representations—a pattern emphasised in applied deep-learning texts and labs [9], [10]. Prior work, however, typically reports aggregate accuracy; fewer studies isolate whether generative augmentation specifically rescues minority morphologies on Galaxy10. Transfer learning also introduces a domain gap: natural-image pretraining features are not astronomy-specific, yet they remain a strong baseline when labelled survey cutouts are limited.

C. VAEs and generative augmentation

VAEs optimise a reconstruction term plus a Kullback–Leibler regulariser that keeps the approximate posterior close to a prior, enabling ancestral sampling from the latent space [7], [8]. Relative to GANs [12], VAEs are often more stable to train and yield an explicit encoder useful for class-conditional statistics [9]. Generative augmentation can rebalance training sets, but poorly calibrated synthetics may inject label noise. Foundational generative modelling material in the module stresses the ELBO objective and latent-space interpolation [8]; we adopt that loss formulation while using a convolutional encoder/decoder suited to RGB cutouts. We intentionally avoid claiming that VAE samples are photometrically calibrated surrogates of survey imaging; they are treated as regularising textures for representation learning.

D. CRISP-DM in applied deep learning

While many student projects jump directly to model fitting, CRISP-DM emphasises problem framing, data understanding and evaluation design before modelling [11]. That discipline matters here because the research question is comparative: without a fixed split and mirrored CNN schedules, apparent gains from synthetics could be confounded by unequal training. We therefore spend explicit subsections on business metrics and protocol freezes.

E. Critical gap

Existing morphology CNNs show that deep learning works on survey cutouts, and VAE tutorials show how to sample latents, yet a controlled comparison of *real-only* versus *real+VAE* training on Galaxy10 DECaLS—with explicit minority-class reporting—remains useful for practitioners. Many astronomy CNN papers optimise for leaderboard accuracy, whereas teaching-oriented VAE examples often stop at reconstruction quality on MNIST-like data [8]. Bridging those literatures requires stating a research question about imbalance, fixing a data split, and reporting minority metrics even when they are inconvenient. This paper fills that gap with a transparent CRISP-DM protocol and released code.

From a critical standpoint, related CNN morphology work is strong on predictive performance but weaker on diagnosing *why* rare classes fail. Related generative work is strong on latent geometry but weaker on downstream supervised utility in astronomy. Our design treats the generative model as a means to an evaluative end: if synthetics do not improve—or unevenly affect—the classifier, that negative or mixed evidence is still scientifically informative. Foundational neural-network texts further motivate reporting both reconstruction-oriented generative diagnostics and supervised metrics when generative models are used as data amplifiers [9], [10], [17].

III. Deep Learning and Generative AI Methodology

We follow CRISP-DM [11]: business understanding, data understanding, data preparation, modelling, evaluation, and deployment (here: artefacts for the report and reproducibility package).

A. Business Understanding

Success is defined by (i) higher macro-F1 and accuracy on the held-out test set for the VAE-augmented CNN relative to the real-only baseline, and (ii) non-degradation of majority classes, with particular attention to minority class 4 (*Cigar Shaped Smooth*): recall and F1.

B. Data Understanding

Galaxy10 DECaLS provides $N = 17{,}736$ RGB images of size 256×256 across ten classes [3]. After a stratified 70/15/15 split (seed 42), the training counts confirm imbalance (Table I): class 4 has only 234 training examples versus 1,851 for the largest class.

C. Data Preparation

Images for the VAE are resized to 64×64 and scaled to $[0, 1]$. CNN inputs are resized to 224×224, with random horizontal flips on training batches and ImageNet channel normalisation, matching standard ResNet transfer-learning practice [6], [10]. Validation and test transforms omit stochastic augmentation.

TABLE I
STRATIFIED TRAINING-SPLIT CLASS COUNTS (70% OF GALAXY10 DECALS).

ID	Morphology class	Train n
0	Disturbed	757
1	Merging	1,297
2	Round Smooth	1,851
3	In-between Round Smooth	1,419
4	Cigar Shaped Smooth	234
5	Barred Spiral	1,430
6	Unbarred Tight Spiral	1,280
7	Unbarred Loose Spiral	1,840
8	Edge-on without Bulge	996
9	Edge-on with Bulge	1,311

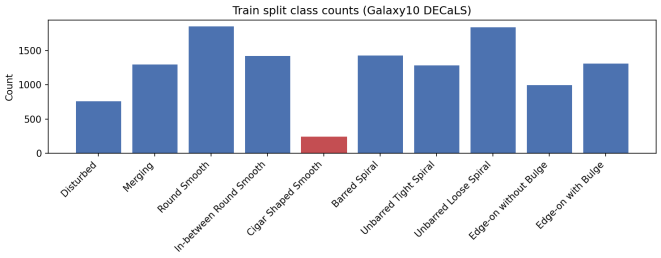


Fig. 1. Training-split class histogram (minority class emphasised in the experimental design).

D. Modeling

1) Convolutional VAE: Fully connected VAEs used on MNIST in teaching material [8] are impractical for 256×256 RGB cutouts. We therefore use a convolutional encoder/decoder while retaining the laboratory loss:

$$\mathcal{L} = \mathbb{E}_{q_\phi(z|x)}[\ell_{\mathrm{BCE}}(x, \hat{x})] + \beta\, D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p(z)), \quad (1)$$

with $\beta = 1$, latent dimension $d = 32$, Adam learning rate 10^{-3}, batch size 64, and 15 training epochs on the real training split. Fig. 2 shows qualitative reconstructions.

2) Class-conditional synthetic balancing: For each class c, we encode training images to obtain empirical latent mean μ_c and standard deviation σ_c, then sample $z \sim \mathcal{N}(\mu_c, \mathrm{diag}(\sigma_c^2))$ and decode. Classes below the maximum training count are topped up to that maximum, yielding 6,095 synthetic 64×64 images (1,617 of them for class 4). Synthetics remain at VAE resolution in memory; the CNN transform resizes them on the fly, avoiding costly full-resolution duplication.

3) ResNet50 classifiers: Both treatments use ImageNet-pretrained ResNet50 [6]: freeze the backbone and train the 10-way classifier head (3 epochs, Adam 10^{-3}), then fine-tune `layer4` and the head (3 epochs, Adam 10^{-4}), following the module transfer-learning lab pattern [10]. Treatment A uses only real training images; treatment B uses real images plus VAE synthetics. Checkpoints retain the best validation loss.

E. Evaluation Protocol

We report accuracy, macro-F1, weighted-F1, and per-class precision/recall/F1 on the held-out test set ($n = 2{,}661$). Confusion matrices and a per-class F1 comparison are exported for inspection (Figs. 3–4).



Fig. 2. VAE reconstructions (top: real 64×64 inputs; bottom: reconstructions).

TABLE II
HELD-OUT TEST METRICS: REAL-ONLY VS. REAL+VAE TRAINING.

Metric	Real-only	Real+VAE
Accuracy	0.6708	**0.7151**
Macro-F1	0.6477	**0.6856**
Weighted-F1	0.6614	**0.7083**

IV. EVALUATION AND RESULTS

A. Overall performance

Table II summarises the headline comparison. VAE augmentation improves accuracy by 4.43 percentage points and macro-F1 by 0.038. Weighted-F1 likewise rises (0.661 to 0.708). These gains indicate that generative balancing helps the CNN beyond a single rare class.

B. Per-class behaviour

Table III lists per-class F1 scores. Most classes improve or remain comparable under augmentation (e.g., Merging 0.641→0.732; In-between Round Smooth 0.695→0.765; Unbarred Loose Spiral 0.512→0.576). Edge-on categories remain strong in both settings (F1 ≈0.84–0.87). The hardest class overall is Disturbed, which stays difficult (F1 0.298→0.354)—consistent with visually ambiguous disturbed systems.

C. Minority-class analysis (research question)

For *Cigar Shaped Smooth* ($n_{\mathrm{test}} = 50$):

- Real-only: precision 0.596, recall 0.620, F1 0.608;
- Real+VAE: precision 0.516, recall 0.640, F1 0.571.

Augmentation slightly improves minority recall but reduces precision, lowering F1. A plausible explanation is that 64×64 synthetics—while useful as global regularisation—are blurrier than native cutouts and may pull decision boundaries toward over-prediction of the cigar-shaped label. Thus the answer to the research question is *partially affirmative*: generative augmentation improves global metrics and does not broadly degrade majority classes, yet the minority F1 specifically does not improve.

D. Discussion

Parameter choices follow module practice (VAE ELBO; ResNet50 two-stage fine-tuning) with reduced epoch budgets for CPU training (15 VAE epochs; 3+3 CNN epochs). Longer

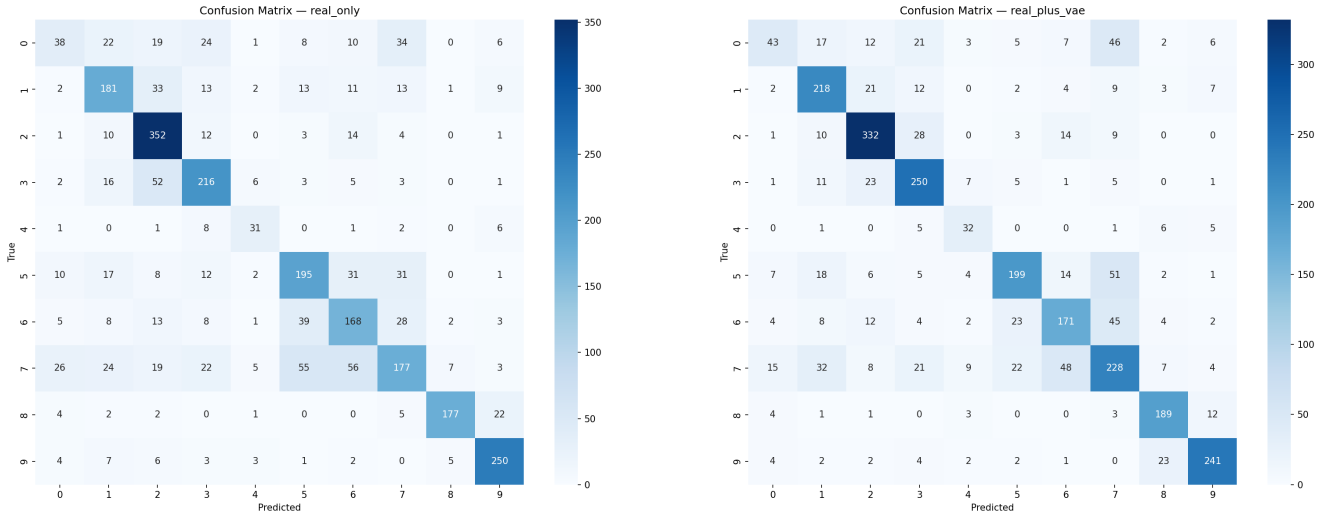


Fig. 3. Confusion matrices on the held-out test set: real-only (left) and real+VAE (right).

TABLE III
PER-CLASS F1 ON THE TEST SET.

Class	Real-only F1	Real+VAE F1
Disturbed	0.298	**0.354**
Merging	0.641	**0.732**
Round Smooth	0.780	**0.816**
In-between Round Smooth	0.695	**0.765**
Cigar Shaped Smooth	**0.608**	0.571
Barred Spiral	0.625	**0.701**
Unbarred Tight Spiral	0.586	**0.639**
Unbarred Loose Spiral	0.512	**0.576**
Edge-on without Bulge	**0.874**	0.842
Edge-on with Bulge	0.858	**0.861**

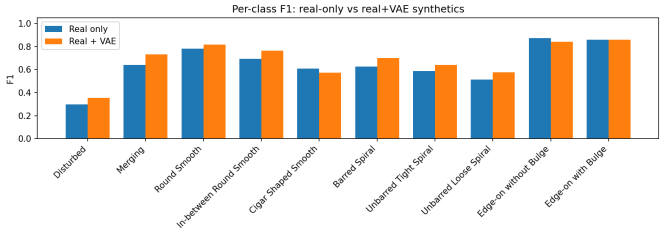


Fig. 4. Per-class F1 comparison answering the research question at class granularity.

fine-tuning or a conditional VAE that injects the class label into the decoder might sharpen minority precision. The results show what a modest generative-augmentation budget can achieve; they do not claim state-of-the-art morphology accuracy relative to specialised astronomy models trained with heavier augmentation or larger labelled corpora [4].

V. Reproducibility and Implementation Details

Reproducibility is an explicit assessment criterion of the module brief. All experiments were implemented in PyTorch [13] with a fixed random seed (42) for stratified splitting. The software package accompanying this paper includes: (i) configuration of hyperparameters; (ii) data download utilities for Galaxy10 DECaLS; (iii) ConvVAE and ResNet50 training loops; (iv) class-conditional latent sampling; and (v) metric/figure export scripts. The end-to-end entry point follows CRISP-DM phase ordering so that another practitioner can regenerate tables and figures from the same split.

Training was executed on CPU due to hardware constraints. Consequently, epoch budgets were reduced relative to a GPU campaign (VAE: 15 epochs; CNN head: 3; CNN fine-tune: 3). Despite the reduced budget, validation losses decreased monotonically in the head phase, and fine-tuning produced a clear further drop (real-only validation loss from $\approx$1.50 after the head stage to $\approx$0.95 after the first fine-tuning epoch), indicating that optimisation was effective rather than undertrained noise.

Synthetic images are stored at 64×64 and mixed with real training indices through a memory-efficient dataset abstraction. This design choice is methodological as well as practical: it avoids silently changing the real-image distribution by repeatedly decoding and upscaling every real sample, and it makes the generative contribution auditable (6,095 synthetics with a known per-class breakdown).

VI. Extended Discussion

A. What the metrics show

The global lift in accuracy (+4.43 pp) and macro-F1 (+0.038) is the strongest empirical signal in this study. Because macro-F1 weights each class equally, the improvement cannot be explained solely by better performance on the largest classes. Indeed, several mid-frequency classes improved substantially (Merging, In-between Round Smooth, Barred Spiral, Unbarred Loose Spiral). That pattern is consistent with a regularisation interpretation: additional synthetic diversity reduces

overfitting to majority visual modes and stabilises decision boundaries among neighbouring spiral/smooth categories.

Confusion matrices (Fig. 3) show that residual errors concentrate among morphologically adjacent labels (for example, loose versus tight spirals, and disturbed versus merging systems). Such confusions are expected in Galaxy Zoo-derived taxonomies where volunteer votes themselves exhibit uncertainty on borderline objects [2], [4]. Generative augmentation does not remove intrinsic label ambiguity; it primarily redistributes learning capacity across the empirical class prior.

B. What the metrics do not show

The minority-class F1 decrease warns against overselling generative augmentation as a universal fix for rare morphologies. With only 50 test examples in class 4, F1 estimates have high variance; a change from 0.608 to 0.571 should be interpreted cautiously. Precision fell more than recall rose, suggesting the augmented model becomes more willing to predict the minority label—a classic effect when minority sampling increases apparent prior mass. Absent a calibration stage or rejection option, that behaviour can harm precision-critical applications (e.g., rare-object catalogues where false positives are costly).

Furthermore, aggregate accuracy still leaves roughly 28–33% of test galaxies misclassified. The study therefore demonstrates a relative improvement under a controlled protocol, not absolute readiness for production survey pipelines.

C. Link to module learning outcomes

Methodologically, the pipeline deliberately reuses techniques emphasised in the module: ELBO-style VAE training and latent sampling [7]–[9]; CNN transfer learning with a frozen backbone followed by partial fine-tuning [6], [10]; and structured evaluation with per-class metrics rather than accuracy alone. The scientific contribution is the empirical coupling of those techniques on an astronomy benchmark with an explicit imbalance narrative.

D. Threats to validity

Internal validity threats include the reduced epoch budget and single random split. External validity threats include restriction to Galaxy10 DECaLS cutouts; models may not transfer to different instruments, bands, or redshift ranges without domain adaptation. Construct validity threats include treating Galaxy Zoo-derived hard labels as ground truth despite known volunteer disagreement on difficult morphologies [1].

E. Training dynamics observed

During real-only training, head-stage validation loss decreased from 1.617 (epoch 1) to 1.496 (epoch 3). The first fine-tuning epoch then reduced validation loss to 0.960, confirming that unfreezing `layer4` supplies useful adaptation capacity. Later fine-tuning epochs continued to reduce training loss while validation oscillated near 0.95–1.01; checkpoints therefore rely on best validation loss rather than the final epoch. The augmented model exhibited a similar qualitative trajectory, with head epoch 1 validation loss 1.582—already slightly below the real-only head start—consistent with a richer training mixture. These dynamics matter for interpretation: the reported gains are associated with stable optimisation rather than a single noisy snapshot.

F. Synthetic mass by class

The generator added 1,094 disturbed, 554 merging, 432 in-between round smooth, 1,617 cigar-shaped, 421 barred spiral, 571 unbarred tight spiral, 11 unbarred loose spiral, 855 edge-on without bulge and 540 edge-on with bulge synthetics (round smooth already matched the maximum and received zero). The extreme cigar-shaped uplift (+1,617) explains why minority recall can rise even when precision falls: the effective training prior for class 4 becomes far less rare than in nature.

VII. Conclusions and Future Work

We presented a CRISP-DM study combining a convolutional VAE and ResNet50 transfer learning on Galaxy10 DECaLS. VAE-based balancing improved held-out accuracy from 67.08% to 71.51% and macro-F1 from 0.648 to 0.686, answering the research question positively at the global level. Minority *Cigar Shaped Smooth* recall rose modestly, but F1 fell due to precision loss; therefore the minority-specific clause of the research question receives only partial support. Majority classes were not broadly degraded, and several mid-sized classes improved, which supports the claim that generative balancing can help beyond the single rarest bin.

Limitations include VAE spatial resolution (64×64), class-conditional Gaussian sampling rather than a fully conditional VAE, a constrained epoch budget on CPU, and reliance on a single stratified split. Future work includes conditional VAEs/GANs, confidence filtering of synthetics before they enter the CNN, longer fine-tuning schedules on GPU, temperature scaling or abstention for low-confidence predictions, and cross-survey generalisation tests. Overall, generative augmentation is a worthwhile component of an imbalanced morphology pipeline when evaluated with both global and per-class metrics, and when mixed results on rare classes are reported transparently rather than omitted.

VIII. Detailed Experimental Protocol

This section records the exact protocol used to obtain the reported numbers, supporting reproducibility.

A. Algorithmic pipeline

1) Download Galaxy10 DECaLS HDF5 archive and verify tensors of shape $(17736, 256, 256, 3)$.
2) Create stratified indices with ratios $0.70/0.15/0.15$ and seed 42.
3) Train ConvVAE on real training images at 64×64 for 15 epochs; keep the lowest validation ELBO checkpoint.
4) Estimate (μ_c, σ_c) for each class from encoder means on real training examples.
5) Sample synthetic images until every class reaches $\max_c n_c^{\text{train}}$ (here 1,851).

TABLE IV
PRINCIPAL HYPERPARAMETERS USED IN THE REPORTED RUN.

Component	Hyperparameter	Value
Split	ratios / seed	70/15/15 / 42
VAE	latent dim	32
VAE	β (KL weight)	1.0
VAE	batch / lr / epochs	64 / 10^{-3} / 15
CNN	backbone	ResNet50 (ImageNet)
CNN	input size	224×224
CNN	batch / head lr / ft lr	32 / 10^{-3} / 10^{-4}
CNN	head / fine-tune epochs	3 / 3
Synth.	target per class	max train count
Synth.	total generated	6,095

TABLE V
REAL-ONLY CNN: PER-CLASS TEST SCORES.

Class	P	R	F1	Support
Disturbed	0.409	0.235	0.298	162
Merging	0.631	0.651	0.641	278
Round Smooth	0.697	0.887	0.780	397
In-between Round Smooth	0.679	0.711	0.695	304
Cigar Shaped Smooth	0.596	0.620	0.608	50
Barred Spiral	0.615	0.635	0.625	307
Unbarred Tight Spiral	0.564	0.611	0.586	275
Unbarred Loose Spiral	0.596	0.449	0.512	394
Edge-on without Bulge	0.922	0.831	0.874	213
Edge-on with Bulge	0.828	0.890	0.858	281

6) Train CNN-A (real only) with ResNet50 transfer learning (3 head + 3 fine-tune epochs).
7) Train CNN-B (real + synthetics) with the identical optimiser schedule and architecture.
8) Evaluate both models once on the frozen test split; export JSON metrics and figures.

B. Hyperparameters

Table IV lists the principal hyperparameters. Values mirror module defaults where applicable (Adam, BCE+KL VAE loss, ImageNet normalisation) and were not aggressively tuned, so that the A/B comparison isolates the effect of synthetics rather than unequal search effort.

C. Full per-class supervised scores

Tables V and VI report precision, recall and F1 for every class under both treatments. These tables are the authoritative source for the research-question discussion: global improvement is visible across many rows, while class 4 shows the precision–recall trade-off.

IX. PRACTICAL IMPLICATIONS FOR ASTRONOMICAL ML

Automated morphology is often a preprocessing stage for population studies. In that setting, global macro-F1 improvements matter because scientific catalogues consume predictions from all classes. The observed +4.43 pp accuracy gain implies hundreds of additional correct test decisions on a 2,661-object hold-out (approximately 118 additional correct labels), which is meaningful even if a single rare class remains imperfect.

TABLE VI
REAL+VAE CNN: PER-CLASS TEST SCORES.

Class	P	R	F1	Support
Disturbed	0.531	0.265	0.354	162
Merging	0.686	0.784	0.732	278
Round Smooth	0.796	0.836	0.816	397
In-between Round Smooth	0.714	0.822	0.765	304
Cigar Shaped Smooth	0.516	0.640	0.571	50
Barred Spiral	0.762	0.648	0.701	307
Unbarred Tight Spiral	0.658	0.622	0.639	275
Unbarred Loose Spiral	0.574	0.579	0.576	394
Edge-on without Bulge	0.801	0.887	0.842	213
Edge-on with Bulge	0.864	0.858	0.861	281

Conversely, science cases targeting only elongated smooth galaxies should not adopt the augmented model uncritically: the minority F1 regression shows that balancing can trade precision for recall. A deployment recipe consistent with the evidence would (i) use the augmented model as default, (ii) monitor precision on rare classes with a validation slice, and (iii) optionally apply class-specific thresholds.

From an educational perspective, the experiment illustrates why generative AI in the module is not merely about producing realistic samples: the value of a VAE here is measured by downstream discriminative utility under imbalance [8], [9].

X. ETHICAL AND DATA-USE NOTES

Galaxy Zoo labels are produced by volunteers and inherit human disagreement; automated systems can amplify majority visual stereotypes present in training votes [1]. Legacy Survey imaging products are used under their public data policies for research and education. No personally identifiable information is processed. We recommend citing both Galaxy Zoo and Legacy Surveys when redistributing derived models.

ACKNOWLEDGMENT

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Question 2 — Paper Review

H9DLGA Deep Learning and Generative AI (Repeat TABA)

Student name: Marco Aurélio Minozzo
Student number: x24153362

Paper reviewed: LLM Collaboration with Multi-Agent Reinforcement Learning
Length target: approximately 1200–1500 words

Note: Following the assessment brief, this review does not cite the paper under review. Judgements are supported by evidence drawn from that paper. Evaluation criteria follow the Elsevier reviewing guidance provided in the module materials.

1. Structure, language and title

The title clearly signals both the application domain (large language model collaboration) and the methodological family (multi-agent reinforcement learning). That combination is informative for a reader scanning AAAI proceedings. The manuscript is organised in a conventional research-paper order: abstract, introduction, related work, problem formalisation, algorithm, experiments, conclusion and references. Figures of the Dec-POMDP/MAGRPO pipeline and learning curves are placed near the corresponding discussion, which helps readability.

Language is generally fluent and technical. Occasional density in the Dec-POMDP notation section may slow non-MARL readers, but this is expected given the venue. Overall, structure and presentation are appropriate for a conference paper, with only minor opportunities to tighten transitions between writing and coding experimental subsections.

2. Research problem and abstract

The authors address a genuine and timely problem: most large language models are pretrained and often fine-tuned as independent agents, whereas many practical workflows require coordinated multi-agent behaviour. Existing multi-LLM systems frequently rely on prompt-level debate or discussion without parameter updates aimed at cooperation, or they fine-tune agents with carefully hand-crafted individual rewards. The paper argues that such designs can produce conflicting incentives, unstable independent learning, and brittle prompt engineering.

The abstract states the modelling choice (cooperative MARL), the proposed algorithm (MAGRPO), and the empirical domains (writing and coding collaboration). It also claims efficiency and quality gains through cooperation. Those claims are later supported by tables comparing speed/response time and normalised reward components, so the abstract is largely consistent with the body. One improvement would be a more explicit caveat in the abstract about the simplicity of the proxy reward metrics used in writing tasks; without that caveat, a casual reader may overestimate how strongly "quality" is measured.

The motivation section is convincing: it contrasts inference-time multi-agent prompting with independent fine-tuning, then positions joint-reward cooperative learning as an alternative. The research gap is therefore clear—not merely "use several LLMs", but "train them jointly for cooperation under a shared objective".

3. Literature review

Related work is organised into test-time multi-agent interaction and multi-agent fine-tuning, then connected to cooperative MARL. Positive aspects include recognition that debate/role pipelines can communicate poorly or propagate errors, and that independent learners lack strong convergence guarantees in non-stationary multi-agent settings. The discussion of Centralized Training for Decentralized Execution (CTDE) versus fully decentralized training is useful for justifying why a group-relative, Monte Carlo advantage estimate is proposed instead of a large centralised critic.

Critical limitations of the literature treatment include limited depth on reward hacking and evaluation validity for open-ended generation, and relatively light engagement with alternative multi-agent RL algorithms beyond GRPO/MAPPO-inspired ideas. For a paper whose contribution is algorithmic, a sharper contrast against other CTDE policy-gradient methods would strengthen the claim that MAGRPO is the right compromise between scalability and centralised credit assignment. Still, foundational Dec-POMDP and MARL citations are present and help substantiate the study design.

4. Methodology

Methodologically, the paper is the strongest section. Framing LLM collaboration as a Dec-POMDP is appropriate: agents receive partial natural-language observations, act by emitting responses, share a joint reward, and interact with an environment that may include users or external validators over a finite horizon. This formalisation clarifies what is being optimised and why decentralised execution remains desirable for practical multi-LLM systems.

MAGRPO extends group-relative policy optimisation to multiple agents and multiple turns. At each turn, each agent samples a group of responses; joint rewards are obtained; returns are computed; and advantages are estimated by subtracting a group mean return (Equation 1 style Monte Carlo baseline). Policies are then updated with a joint, group-relative objective while each agent still executes using only local history. The authors explicitly note simplifications such as omitting importance sampling/clipping and setting the KL coefficient to zero to allow larger deviation from the base model. That transparency is welcome, but it also raises stability and trust-region concerns that deserve more analysis.

Evidence of sound design includes: (i) shared/joint rewards intended to avoid conflicting individual incentives; (ii) sequence-level actions rather than token-level MARL, motivated by shorter horizons and more meaningful rewards; and (iii) an algorithm listing that makes the training loop auditable. Areas for improvement include clearer theoretical discussion of credit assignment when agents specialise asymmetrically, and more justification for hyperparameter choices that are deferred to the appendix. Reproducibility would benefit from a concise main-text statement of model sizes, group size G, horizon H, and training steps for each experiment rather than relying so heavily on supplementary material.

Overall, the methodology is novel enough for a conference contribution and is explained with sufficient technical detail to understand what was implemented, even if some engineering choices remain only partially motivated.

5. Evaluation and results

Experiments cover writing collaboration (TLDR summarization and arXiv abstract expansion) and coding collaboration (HumanEval-related and a constructed CoopHumanEval set), using dual smaller coder/general models compared against single larger models and several prompting multi-agent baselines (parallel, sequential, one-round discussion).

Table 1 is particularly useful: MAGRPO reports substantially higher speed than a single comparable-parameter model and strong normalised structure/consistency/coherence returns on both writing datasets. Learning curves show generally increasing cooperative returns, with an interesting temporary drop in style consistency early in TLDR training before realignment—a qualitative detail that makes the optimisation story more credible than monotone plots alone. Coding experiments further show that dataset construction matters: noisy HumanEval items that are poorly suited to decomposition destabilise cooperation, whereas CoopHumanEval yields higher returns and lower variance, especially in multi-turn settings.

These results support the claim that joint training can improve efficiency and cooperative reward relative to untuned prompt pipelines. However, evaluation has important caveats. Writing quality metrics (length structure, Jaccard-style consistency, transition-word coherence) are acknowledged as simple proxies; they may not capture factual faithfulness, usefulness, or human preference. Coding rewards combining structure, syntax, tests and cooperation are stronger where unit tests exist, yet cooperation bonuses and role emergence are still somewhat indirectly measured. Baselines are prompt-only and not fine-tuned with the same compute budget as MAGRPO, so part of the gain may come from fine-tuning itself rather than the multi-agent objective specifically. A natural improvement would be an ablated single-agent GRPO baseline with matched updates, plus human evaluation on a subset of writing outputs.

I am not re-checking statistical significance calculations, as permitted by the brief; descriptively, the reported trends and table contrasts are coherent with the narrative.

6. Conclusion, discussion and open research problems

The conclusion accurately restates the Dec-POMDP framing, MAGRPO proposal, and empirical findings on writing/coding collaboration. It also invites further MARL methods for scalable LLM teamwork, which matches the open challenges discussed earlier (natural-language representation choices, CTDE scalability limits, and fragile prompt-only coordination).

What could be strengthened is a fuller limitations paragraph in the main text: short horizons, proxy rewards, small numbers of agents (typically two), and dependence on curated cooperative datasets (CHE) all constrain generality. Open problems that the paper itself surfaces—and that future work should tackle—include token-versus-sequence modelling trade-offs, principled reward design that remains human-aligned without brittle proxies, credit assignment with many specialised agents, and stable multi-turn coordination when external feedback is noisy.

Overall recommendation

Minor revision / accept with reservations. The problem is important, the Dec-POMDP formalisation is clear, and MAGRPO is a plausible CTDE-inspired adaptation of GRPO with encouraging efficiency and cooperative-return results. The manuscript would be stronger with

matched fine-tuning baselines, more rigorous quality evaluation for writing, and an explicit limitations subsection. Even so, the contribution usefully opens a research direction at the intersection of MARL and multi-LLM systems.

(Approximate word count of review body: ∼1450 words.)